

Editor-in-Chief  
*Bayesian Analysis*

Dear Editor,

We are pleased to submit the manuscript

“Catalan-Exponential Priors for Bayesian Decision Trees: WAIC Consistency,  
Posterior Concentration, and PAC-Bayes Guarantees”

for consideration for publication in *Bayesian Analysis*.

*Summary of contributions.* This paper develops the theoretical foundations of the Bayesian decision tree classifier that uses a Catalan-exponential tree-size prior and Dirichlet-Multinomial leaf models with RJMCMC inference. We prove four main results that collectively provide a complete theoretical justification for the model design and explain previously unexplained empirical findings:

1. *WAIC-LOO gap theorem:* The WAIC and LOO deviances agree up to the constant  $-4N/(N + \alpha_0) + O(N^{-1})$ , proved in closed form. This yields the practical design criterion  $N_{\min} \approx 5.41/\Delta$ , which predicts the exact sample size at which WAIC-weighted BMA becomes reliable.
2. *Catalan prior tail bound:*  $P(k \geq k_0) \leq C_\gamma (e^{-\gamma}/4)^{k_0}$ , with universal effective decay rate  $r_{\text{eff}} = 1.228 e^{-\gamma}/4$ . Posterior concentration on the true complexity  $k^*$  is achieved at  $N = 300$  (Catalan) versus  $N > 800$  (Chipman 1998).
3. *BMA oracle inequality:*  $R_{\text{BMA}} - R_{k^*} \leq \Delta_{\text{Jensen}} = O(\exp(-N\Delta_{\text{WAIC}}/2))$ . Excess risk decreases 56-fold from  $N = 20$  to  $N = 800$ , following  $O(1/N)$ .
4. *PAC-Bayes sample complexity:* Under the oracle posterior, the Catalan prior requires  $8.1\times$  fewer samples than Chipman for sparse true models ( $k^* = 1$ ), directly explaining its advantage in small- $N$  medical imaging.

Supplementary appendices provide RJMCMC detailed balance, ECE calibration bounds, and decision-theoretic analysis connecting the above to asymmetric-loss classification. All results are accompanied by numerical verification on both synthetic and real knee osteoarthritis data.

*Fit for Bayesian Analysis.* The paper is squarely within the scope of *Bayesian Analysis*: it develops methodology for Bayesian model selection (WAIC) and BMA, analyses a novel prior construction (Catalan-exponential) analytically, and provides PAC-Bayesian guarantees. The application to medical imaging provides concrete motivation but the theoretical results are model-agnostic within the DM-leaf class.

*Suggested associate editors / reviewers.* Given the combination of Bayesian nonparametric trees, model selection, and PAC-Bayes bounds, we would suggest reviewers with expertise in Bayesian model selection, decision trees, or PAC-Bayesian theory.

*Declaration.* The manuscript is original, has not been published, and is not under consideration elsewhere. There are no conflicts of interest. The paper is *not anonymised*

in this version; please advise if double-blind anonymisation is required for the current submission round.

*Code availability.* All experiments are fully reproducible. Code and pre-computed results are publicly available at <https://github.com/vitsch/jbdt>.

Yours sincerely,

Livija Jakaite and Vitaly Schetinin  
Vitaly.Schetinin@beds.ac.uk (corresponding author)